# Samuel Leblanc

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## Education

**Queen's University** **2025 - 2027**
*M.Sc. in Mathematics*
Research project: Quivers with potential, representation theory, categorification
Supervisors: Ivan Dimitrov, Charles Paquette, and David Wehlau

**Université de Sherbrooke** **2022 - 2025**
*B.Sc. in Mathematics*
Mention of Excellence – Faculty Dean's List, Winter 2025

## Research Interests

Representation theory of (finite-dimensional and associative) algebras, cluster algebras, homological algebra.

## Papers

1. Brüstle, T., Desrochers, J., Leblanc S., *Initial Subcategories via Free Presentations of the Constant Functor.* arXiv:2610.41068 (2026)
2. Brüstle, T., Desrochers, J., Leblanc S., *Generalized Rank via Minimal Subposet.* Linear Algebra and Its Applications 753: 57–81 (2027). arXiv:2510.10837
3. Leblanc, S., Rasolomanana, A., Armenta, M., *Hidden Activations Are Not Enough: A General Approach to Neural Network Predictions.* arXiv:2409.13163 (2024)
4. Armenta, M., Leblanc, S., *Batalin-Vilkovisky structure on Hochschild cohomology with coefficients in the dual algebra.* arXiv:1810.13023 (2018) (I joined as coauthor in 2025; revisions and extensions)

### Student Papers

1. Leblanc, S., *Dégénérations des représentations de carquois de type $\mathbb{A}_3$ à deux puits.* Cahiers mathématiques de l'Université de Sherbrooke 8: 145–167 (2026)
   Supervisors: Emily Cliff and Shiping Liu
2. Baril, L., Leblanc, S., *Multiplicité de la représentation indicatrice.* (2025)
   Supervisor: Thomas Brüstle
3. Leblanc, S., *Transformations de cercles orientés tangents sur la sphère de Riemann.* (2023)
   Supervisor: Jean-Philippe Burelle

## Talks

1. *Triangulations of the $\infty$-gon* (October 23, 2026)
   Grad Seminar, Department of Mathematics and Statistics, Queen's University
2. *(Co)limit Computation and Its Application to Representation Theory* (January 26, 2026)
   Algebra & Geometry Seminar, Queen's University
3. *Generalized Rank in Generalized Linear Algebra* (November 12, 2025)
   Grad Seminar, Department of Mathematics and Statistics, Queen's University
4. *Analyse topologique de données* (February 13, 2025)
   Club mathématiques de l'Université de Sherbrooke
5. *La propagation avant en tant que matrice* (November 14, 2024)
   Club mathématiques de l'Université de Sherbrooke
6. *Visualisation de transformations sur la sphère de Riemann* (March 21, 2024)
   Club mathématiques de l'Université de Sherbrooke
7. *Théorie des représentations des réseaux de neurones* (October 5, 2023)
   Club mathématiques de l'Université de Sherbrooke

### Posters

1. *Minimal Initial Subcategory.* Presented at Queen's Mathematics Summer School and Modern Methods in Representation Theory (2026)
2. *Multiplicity of the Interval Module.* Presented at AARMS-CMS Student Poster Session (CMS Summer Meeting) (2025)
   Collaborators: Laurianne Baril and Justin Desrochers. Supervisor: Thomas Brüstle.
3. *Kernel of the Rank Invariant* (with Justin Desrochers). Presented at Summer Research School, Applications of Representation Theory in Topological Data Analysis and Geometric Invariant Theory (2024)

## Awards & Fellowships

**Senator Frank Carrel Fellowship ($15,000)** **2026 - 2027**
*Queen's University*

**Best Poster Award (Minimal Initial Subcategory)** **2026**
*Queen's Mathematics Summer School*

## Teaching

### Guest Lecturer

**Real Analysis (MATH/MTHE 328)** **2026**
Replaced the course instructor for one week. Delivered 3 hours of lectures.
*Department of Mathematics and Statistics, Queen's University*

### Teaching Assistant

**Algebraic Structures with Applications (MTHE 217)** **2026**
**Differential and Integral Calculus (MATH 121, MATH 128)** **2026**
*Department of Mathematics and Statistics, Queen's University*

**Méthodes quantitatives en communication marketing (MQG301)** **2025**
**Statistique appliquée à la gestion (MQG222)** **2024**
*École de gestion, Université de Sherbrooke*

**Calcul vectoriel (MAT298)** **2024**
*Département de mathématiques, Université de Sherbrooke*

### Grader

**Differential and Integral Calculus, Real Analysis, Algebraic Structures, and Discrete Mathematics.** **2024 - 2026**
*Queen's University* and *Université de Sherbrooke*

## Service

**Academic Coordinator, Queen's GMS** **2026 - 2027**
Coorganizer, with Trevor Shillington, of the Graduate Student Seminar of the Department of Mathematics and Statistics.

**Mathematics Tutor, 10th Grade** **2023 - 2024**
*Volunteering with Le Diplôme avant la Médaille*

## Selected Academic Activities

### Conferences

**Meeting on Representation Theory of Algebras** **2024, 2025 & 2026**
Sherbrooke, QC.

**Route 81 Conference** **2025 & 2026**
Kingston, ON.

**Modern Methods in Representation Theory** **2026**
Ottawa, ON.

**Auslander Conference** **2026**
Woods Hole, MA, USA.

**Canadian Mathematical Society (CMS) Meeting** **2025**
Québec, QC.

### Summer Schools

**Queen's Mathematics Summer School** **2026**
Kingston, ON.

**Summer Research School: Applications of Representation Theory in Topological Data Analysis and Geometric Invariant Theory** **2024**
Montréal, QC.

### Undergraduate Research

**Summer Internship: Representation Theory of Quivers** **2023**
*Département de mathématiques, Université de Sherbrooke*
Supervisors: Emily Cliff and Shiping Liu

## Software

**QPMutation** GitHub
Compute mutations of quivers with potentials.

**knowledgematrix** GitHub
Implement neural networks and compute their knowledge matrices.

**upperhalfplane** GitHub
Visualize the action of $\mathrm{PSL}(2,\mathbb{R})$ on the upper half plane.

**riemannsphere** GitHub
Visualize the action of $\mathrm{PSL}(2,\mathbb{C})$ and $\mathrm{PSp}(4,\mathbb{R})$ on the Riemann sphere.
Supervisor: Jean-Philippe Burelle.

**MatTalX** GitHub
Extension to convert LaTeX commands into plain text.

## Languages

French (native), English (advanced).